Star Formation Rate Density at 2 < z < 7: The Redshift Evolution of Early Galaxies with Hydrogen-Alpha Line Luminosity Functions

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Introduction

- Our Universe's earliest galaxies emitted a lot of high-energy ionizing radiation by forming new stars
- Measuring the star formation rates (SFRs) of early galaxies allows us to model this "Epoch of Reionization"
- Galaxies with H-α spectral emission lines are strong tracers of star formation, so calculating the fluxes of these lines allows us to calculate SFRs
- The James Webb Space Telescope's (JWST) multi-object spectrometer (NIRSpec) enables it to observe H-α at high redshifts, leading to the creation of the RUBIES catalog.[1]
- We present a sample of 504 H-α detected galaxies at redshifts z=2.0-7.0 from the RUBIES JWST/NIRSpec Survey in the EGS and UDS fields

Motivation

Previous studies using H-α to calculate SFR density only reached z~2.[2] Using RUBIES data, we aim to expand this out to z~7, further constraining models of the Universe's distant past.

Data and Methods

- We calculate H-α line fluxes with an MCMC Gaussian fitting algorithm
- We convert these fluxes to luminosities, and create luminosity functions (LFs) by plotting the number density of galaxies at different luminosities
- We completeness correct the LFs by creating a spectral simulation grid using *Pandeia* and cross-matching between NIRSpec and UDS/EGS photometric catalogs[3]
- We fit the LFs to Schechter models with MCMC and integrate them to calculate SFR density

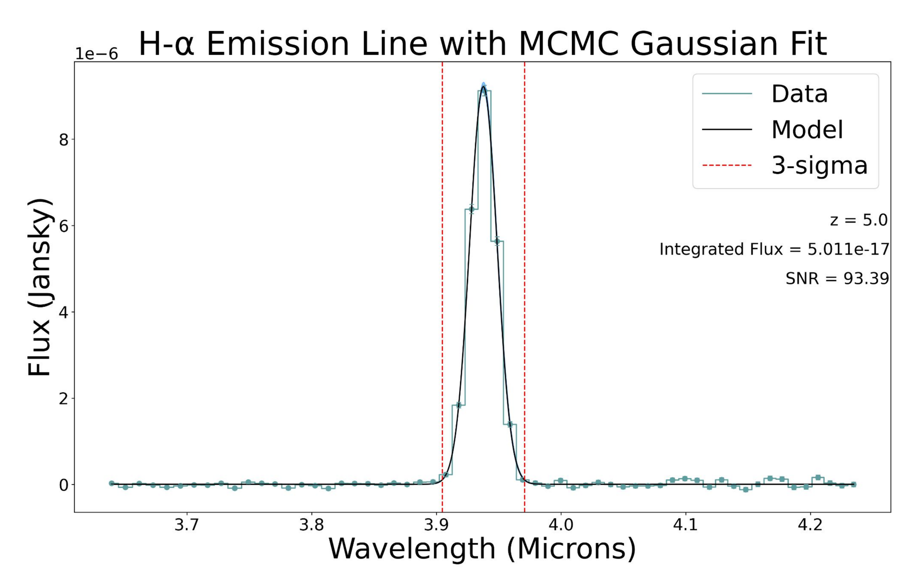


Figure 1. An example H-α spectral emission line fit using an MCMC Gaussian algorithm.

Figures and Results

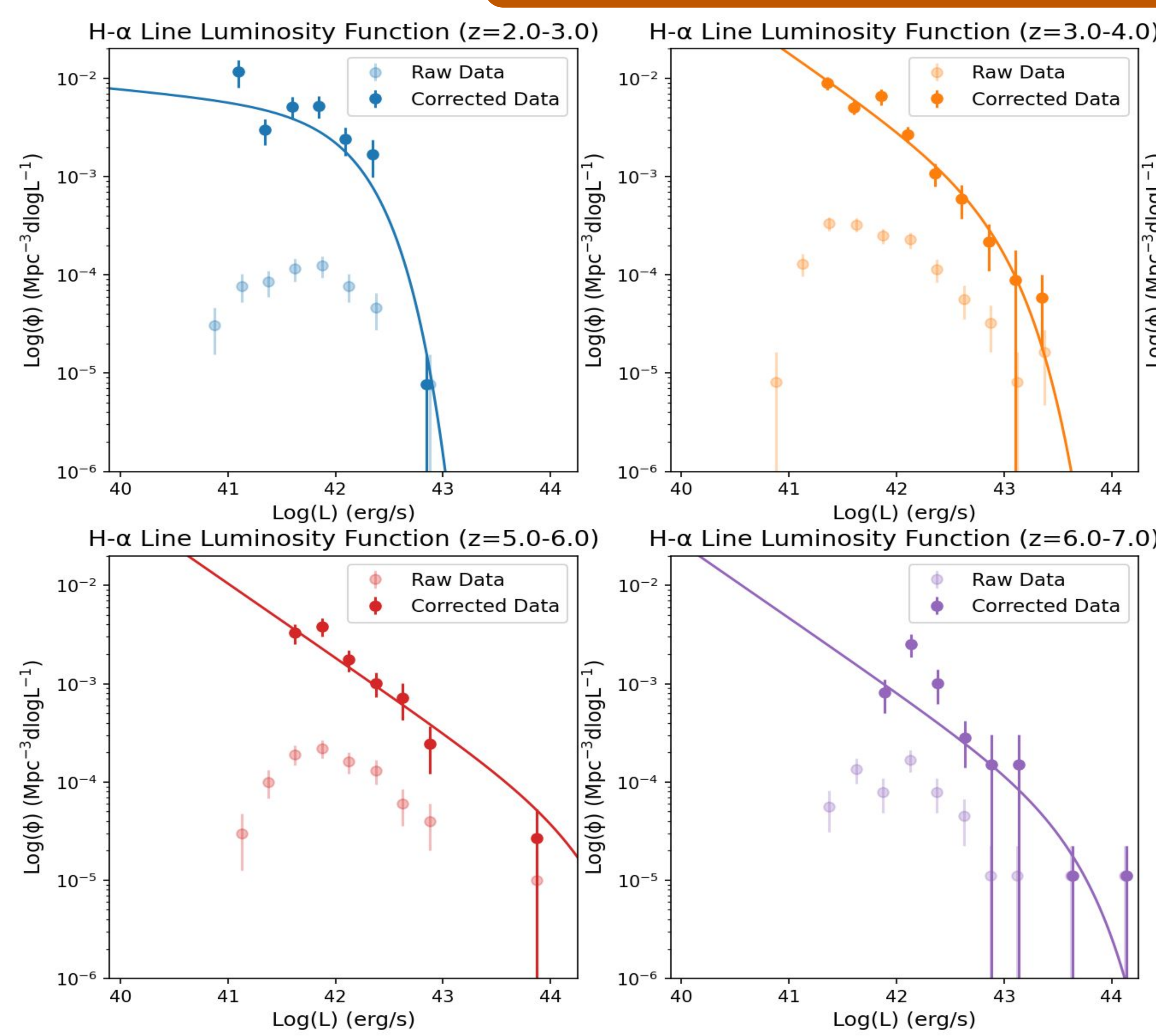
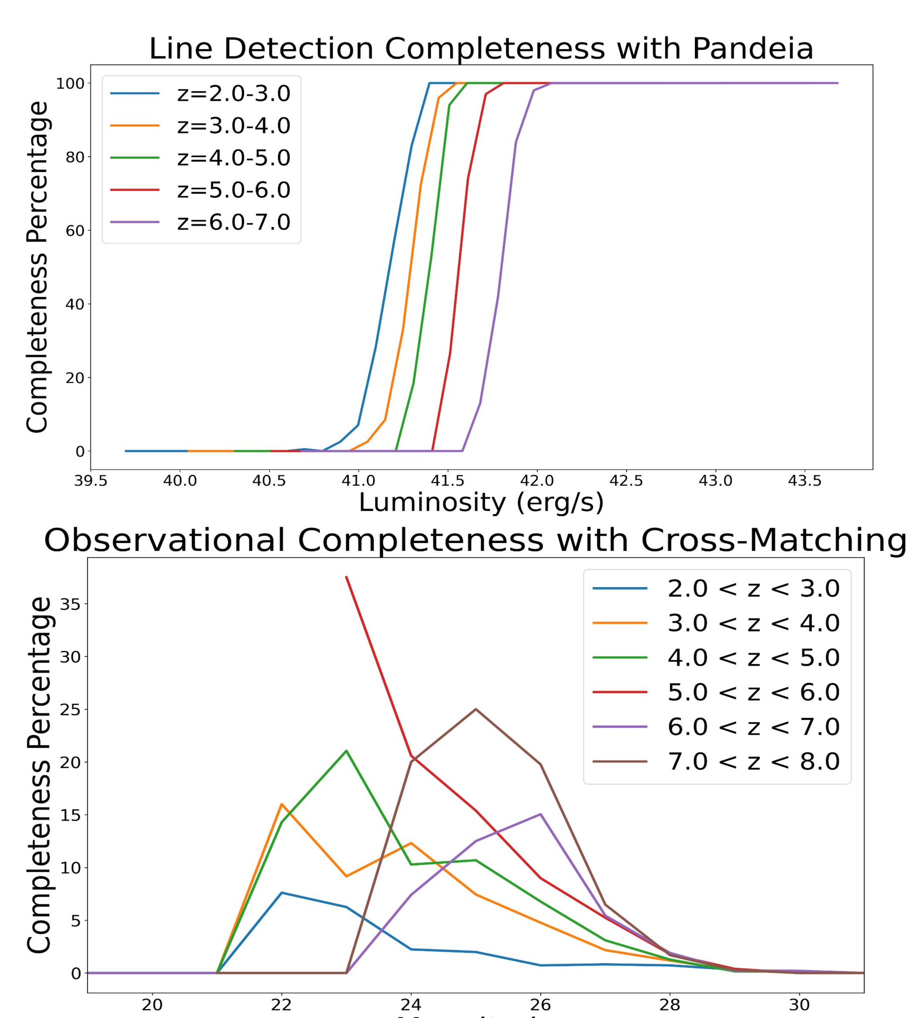


Figure 2. Number of galaxies per MPC are plotted as a function of luminosity at different redshift ranges. The pre-corrected data are shown in faded color shades, while the brighter color shades indicate data that have been corrected for line detection and observational completeness. Schechter fits are applied to each LF using an MCMC algorithm.

Completeness corrections to the H-α LFs account for the MCMC algorithm not detecting all H-α emitters and for NIRSpec not observing spectra for all EGS/UDS objects. This increases the calculated number densities of H-α emitters, thus increasing SFRs. By knowing the SFRs at different redshifts and luminosities, we model how much of the Universe's reionization was due to early galactic star formation, as opposed to active black holes.



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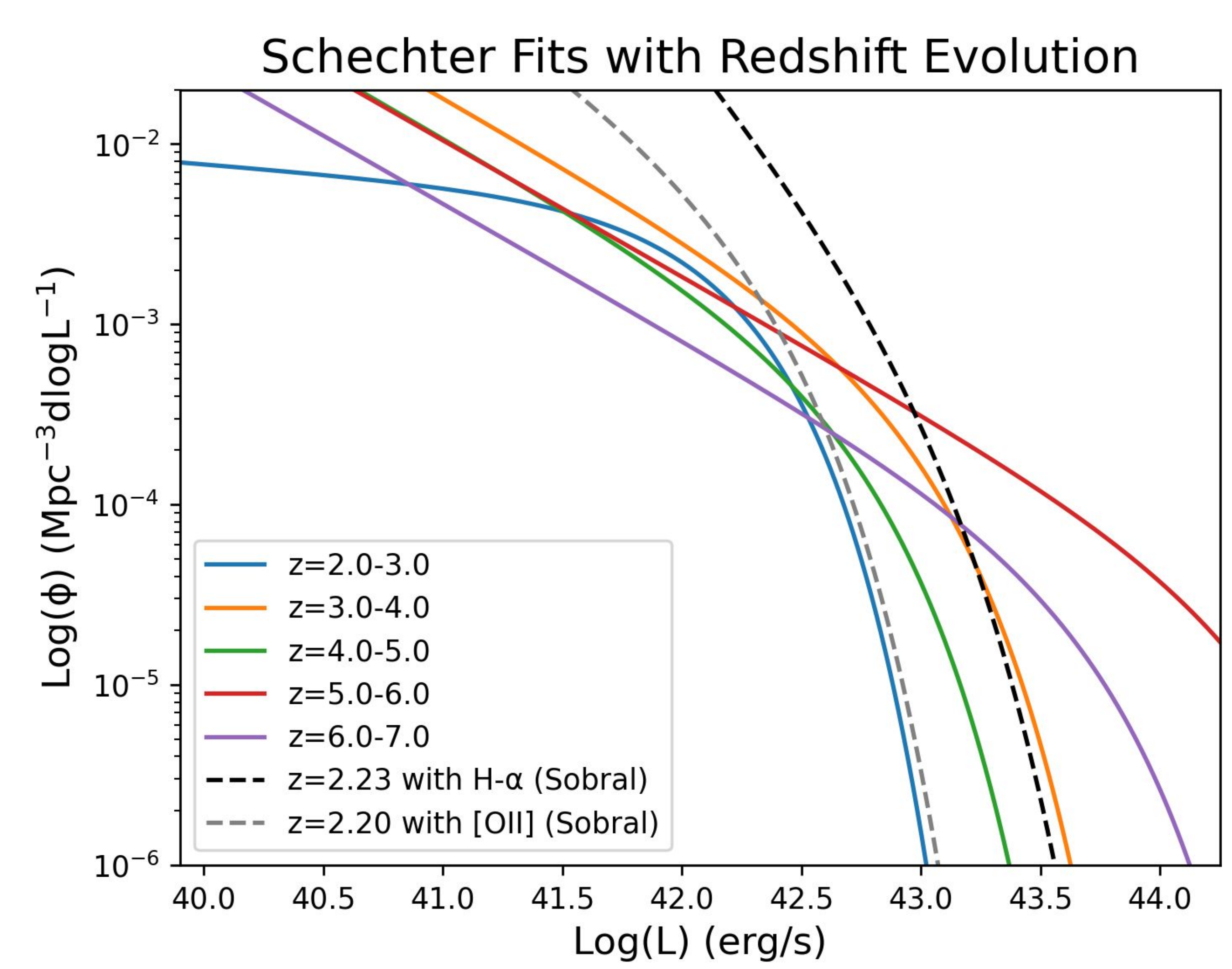

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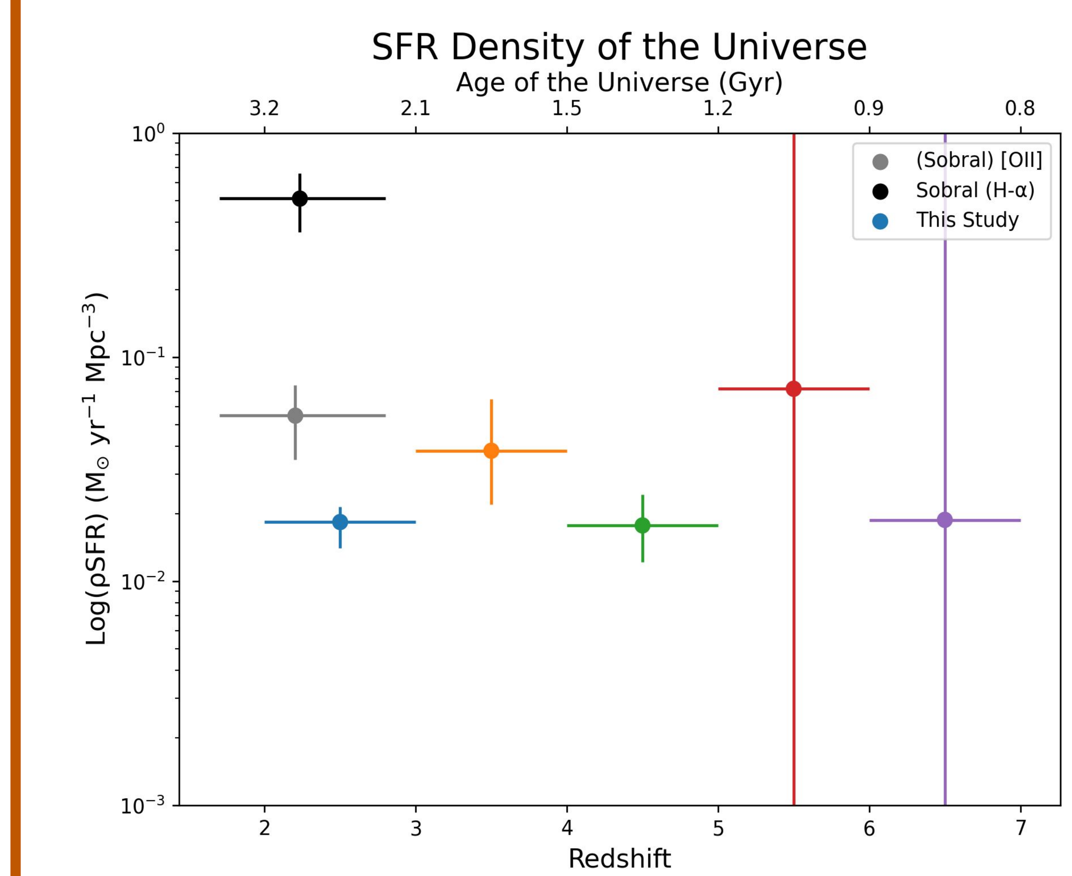
Conclusion

We calculate estimates for the SFR density by integrating the Schechter functions. Measurements up to z=7 expand far beyond previous studies with H-α as a star formation tracer. While we see disagreement between our z=2.0-3.0 bin and Sobral et al. (2015), we can further refine our data by removing active black holes and accounting for the effects of dust in our sample.



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References

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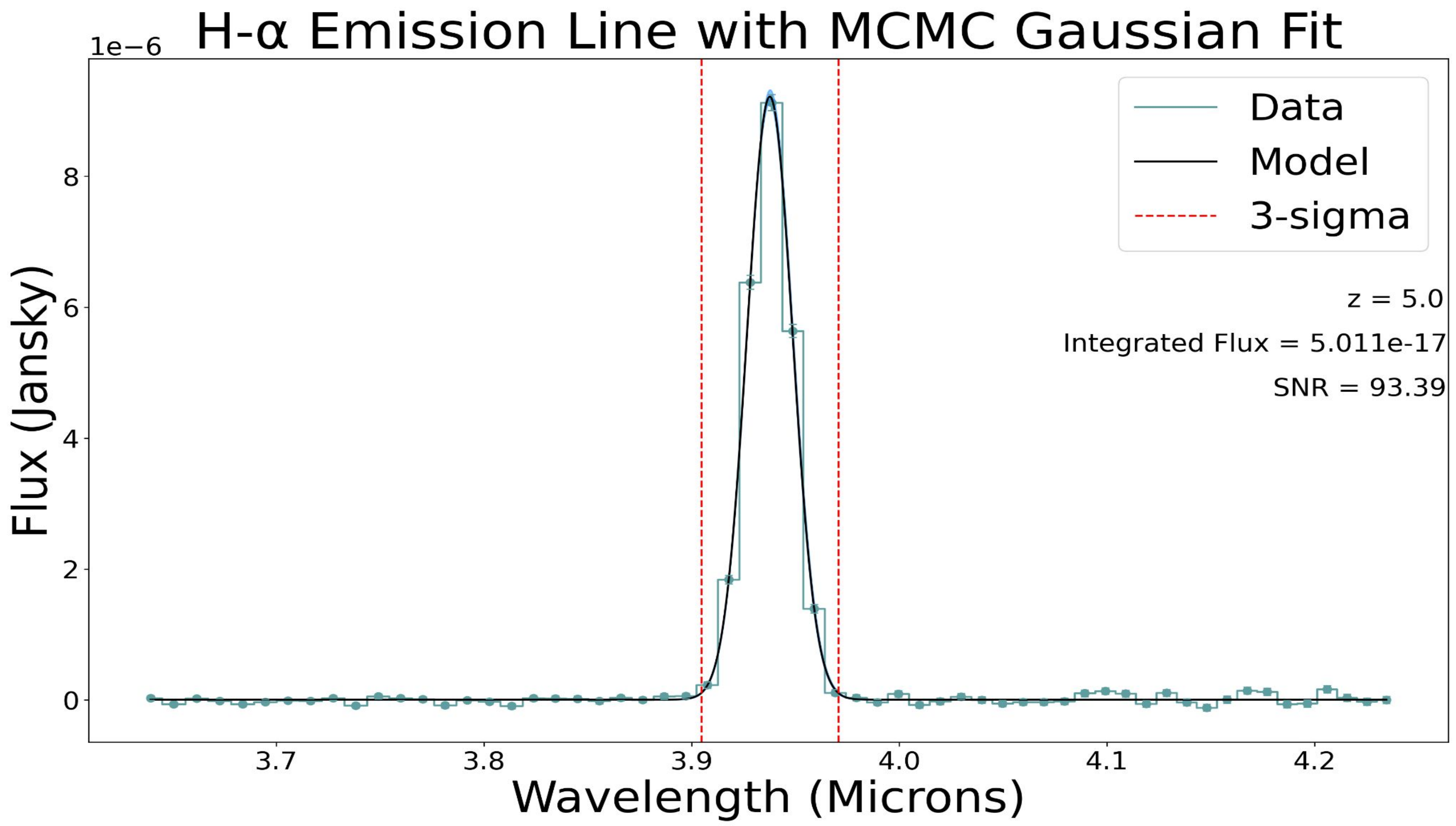

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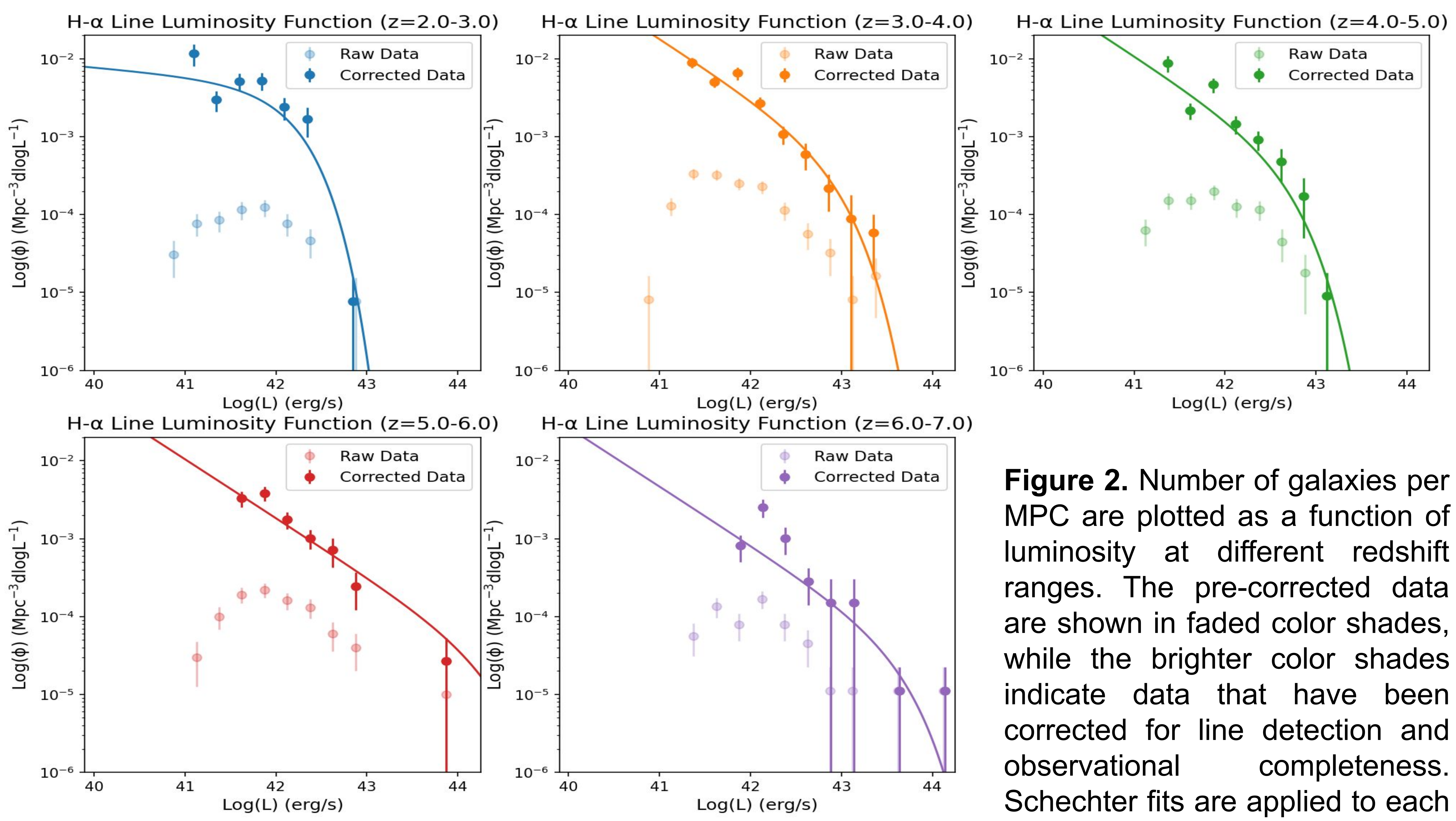

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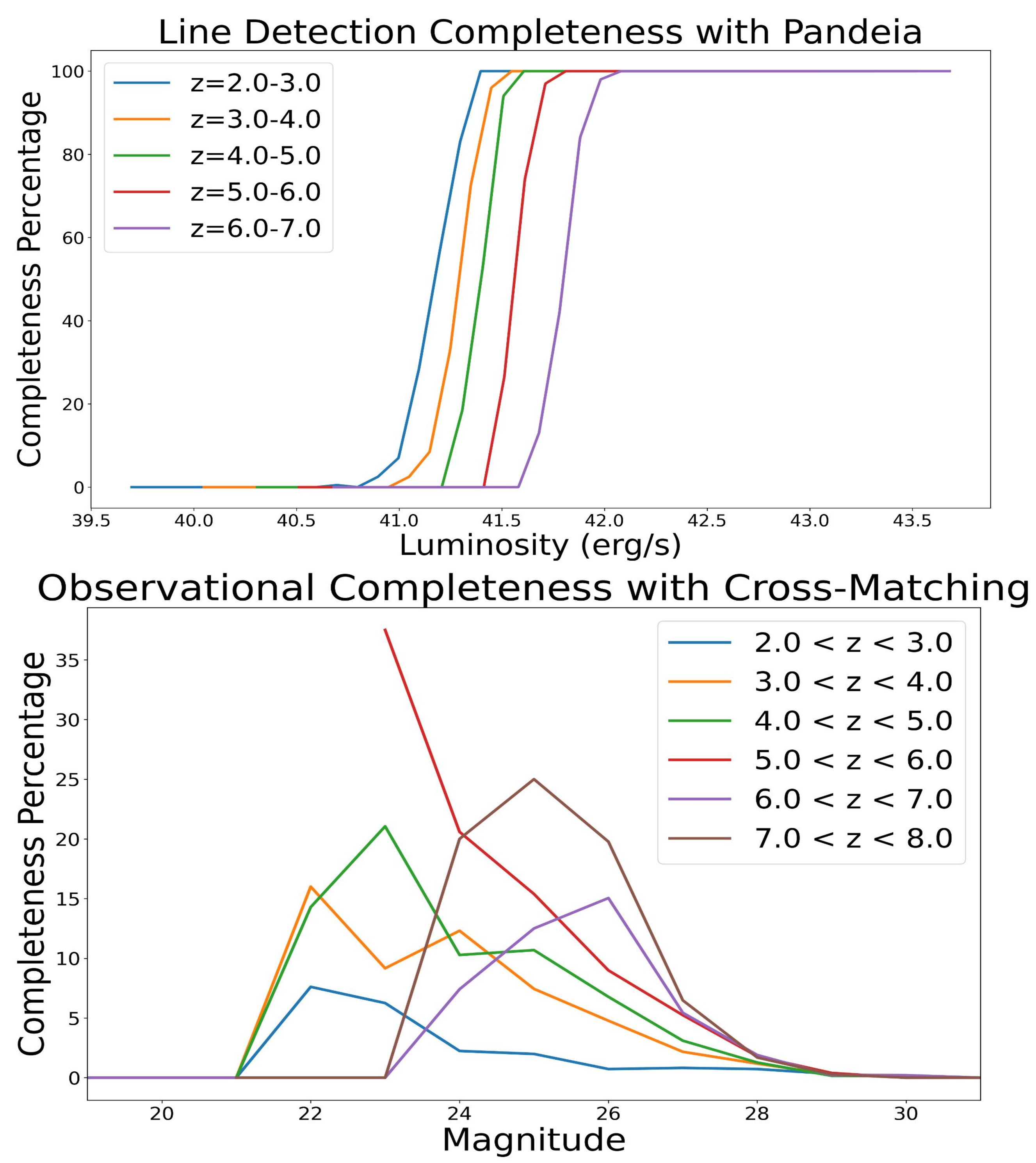

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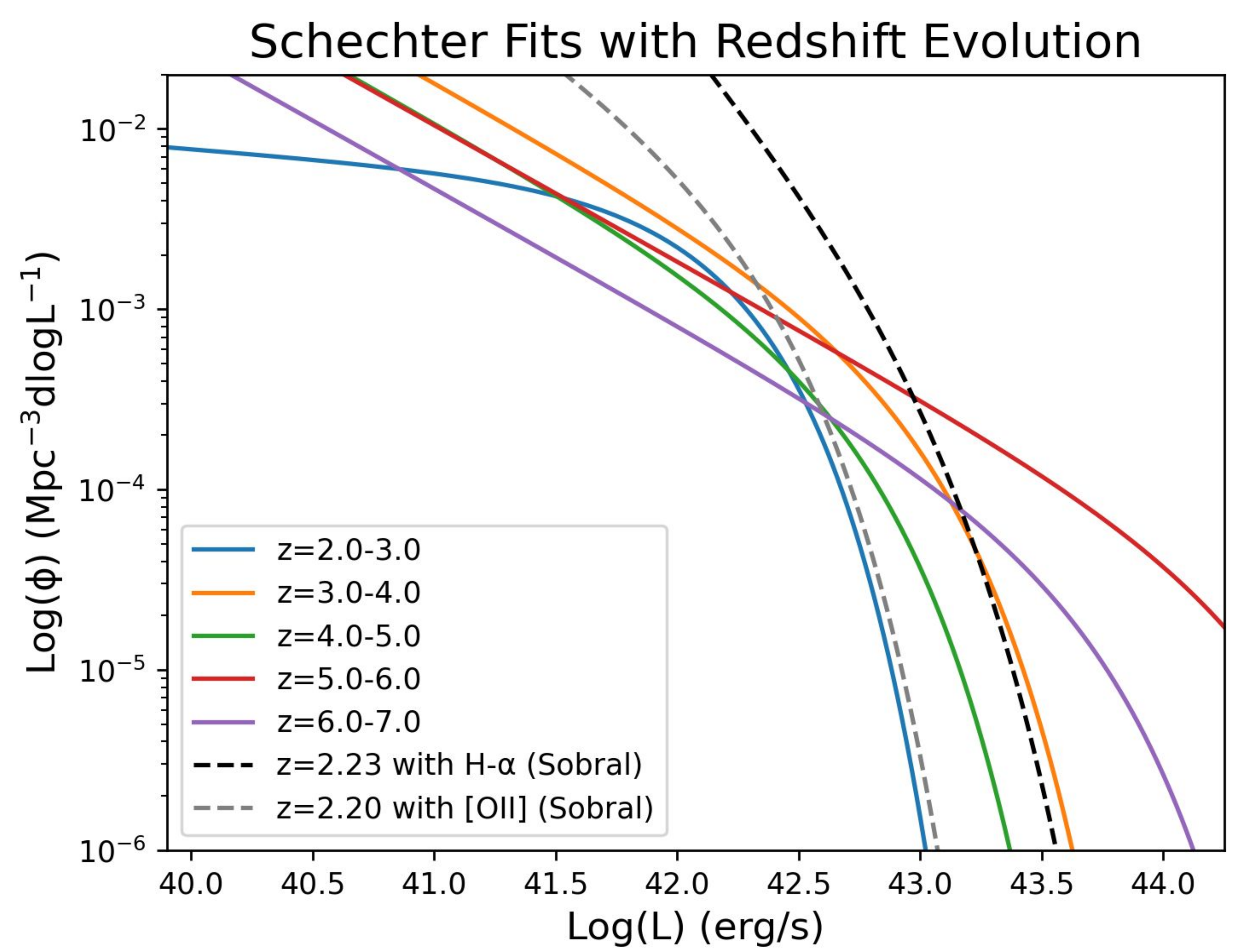

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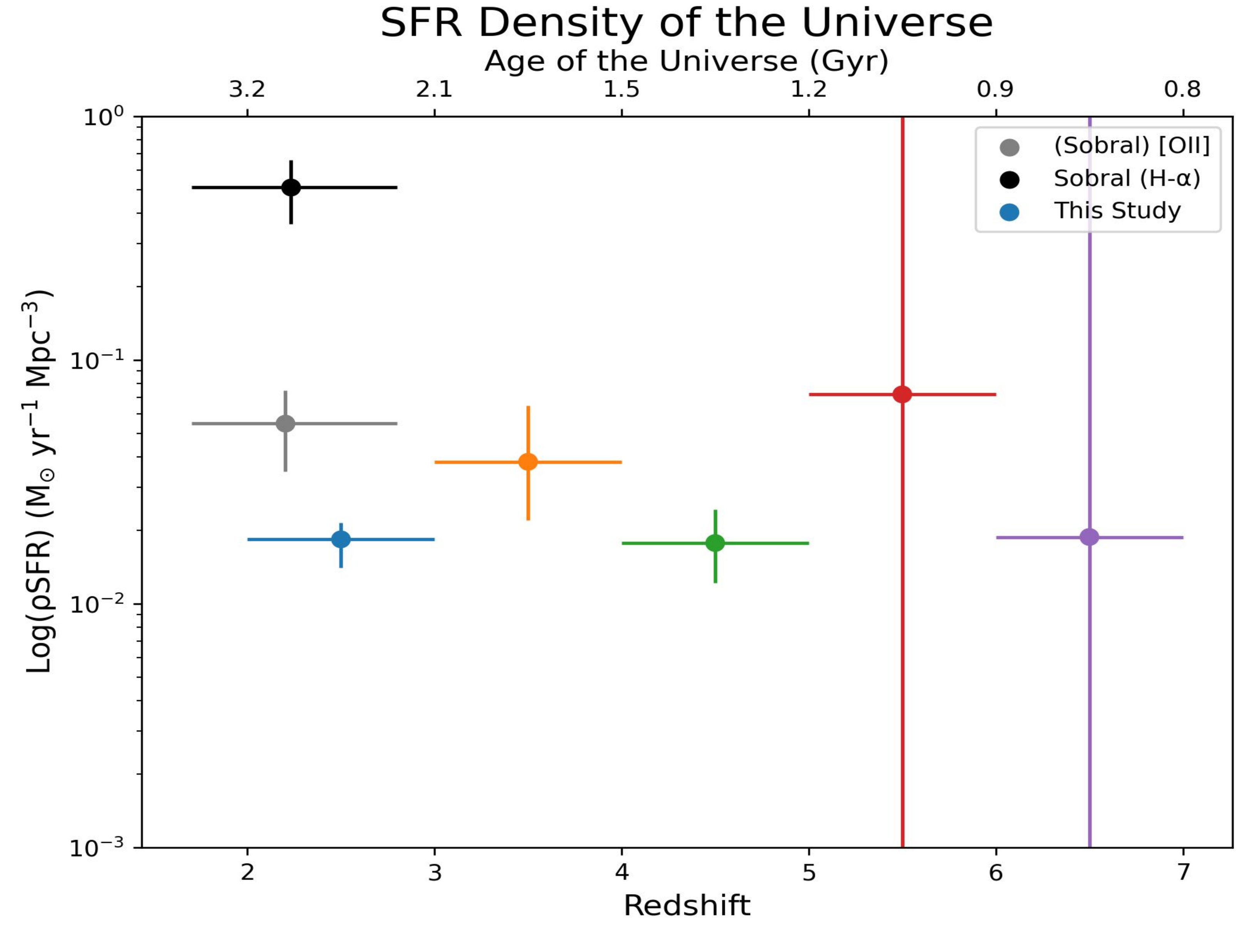

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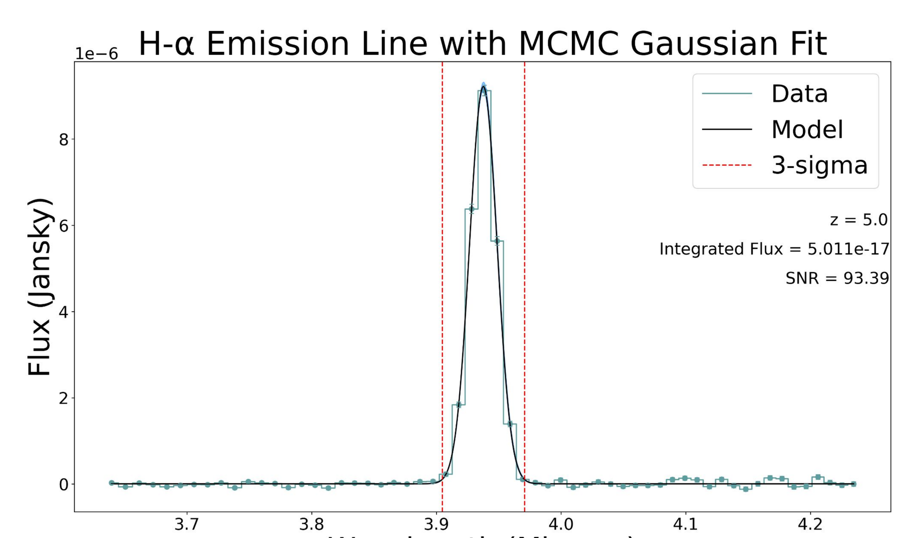

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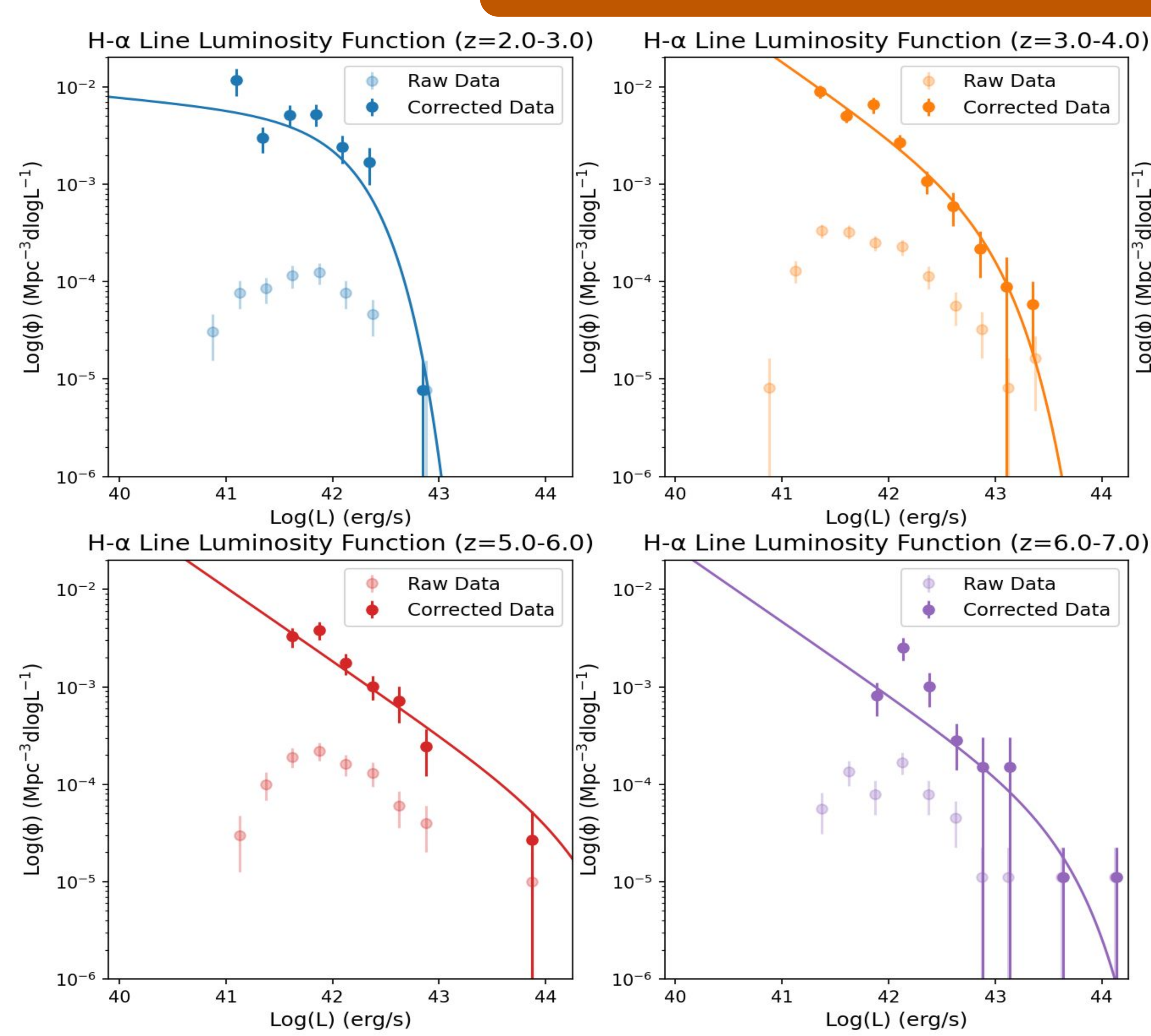

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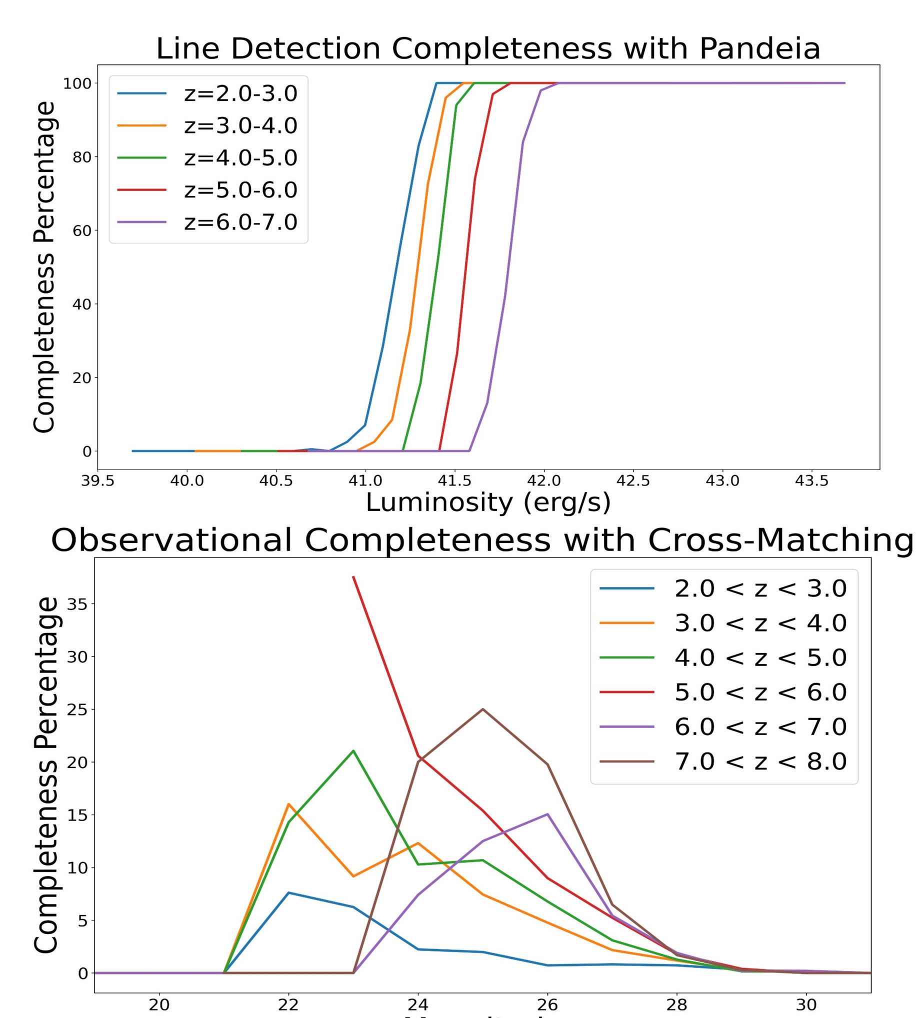

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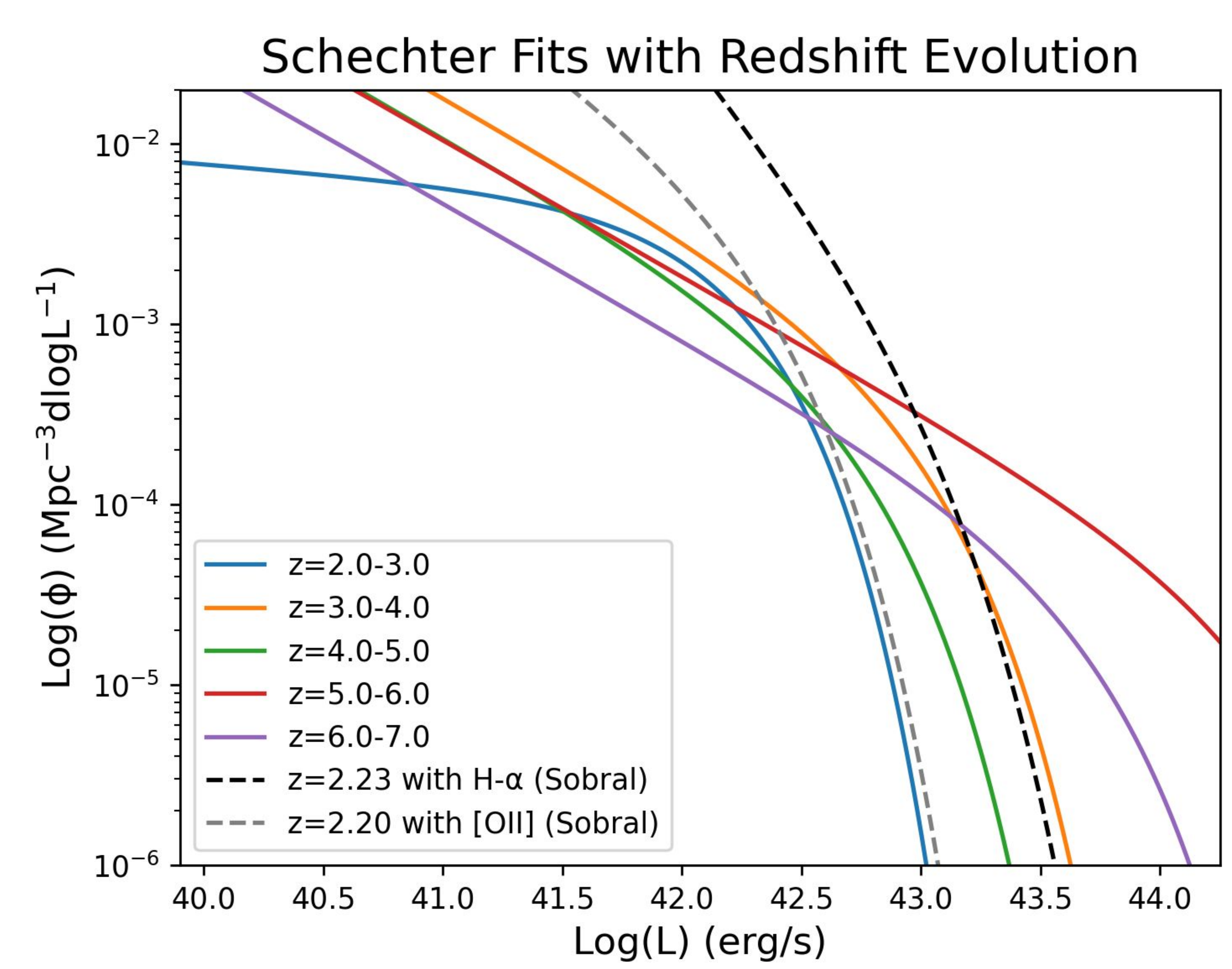

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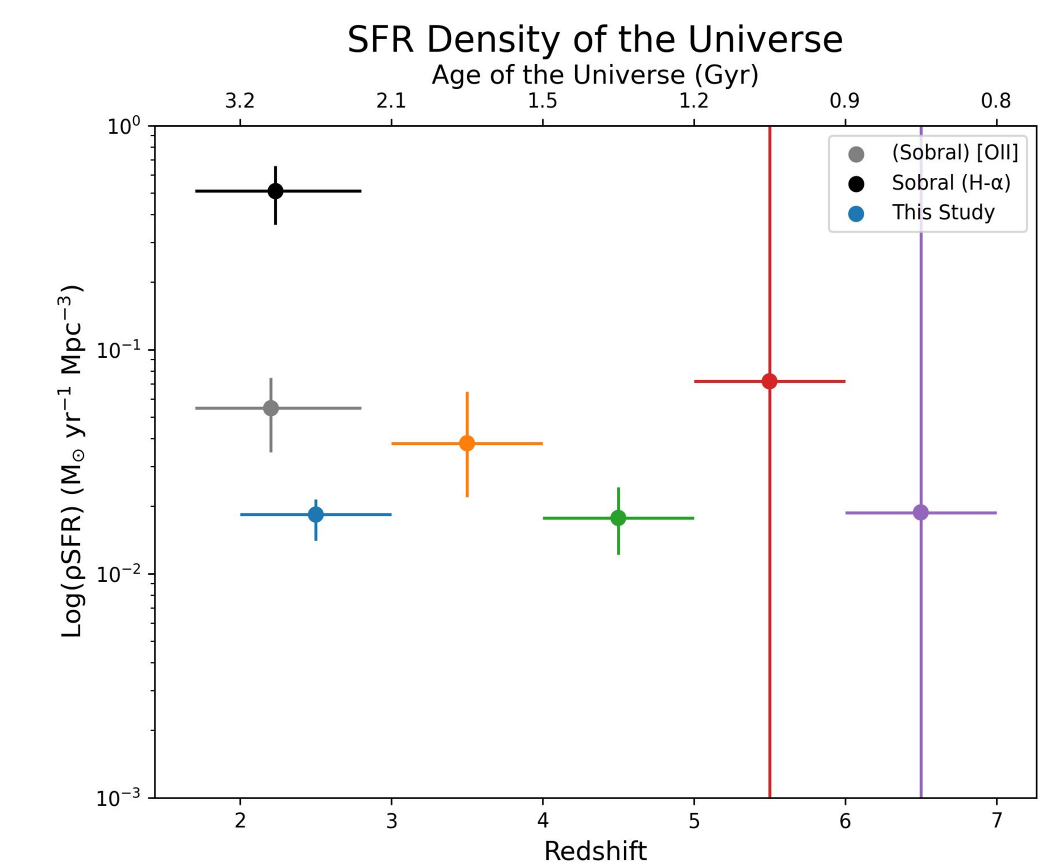

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